vi) Display all name and percentage where percentage is b/w 60 and 70.

Π (name, percentage) = σ {percentage > 60 $\wedge$ percentage < 70} (STUDENT)

Π name = σ Stream $\neq$ 'S' (STUDENT)

45 mks

Exact & Non-Exact ✓
Variable Separation
Homogeneous Eqn. ✓
Linear Eqn ✓
~~Clairaut Eqn.~~ Clairaut's Eqn.
Bernoulli
Order & Degree
Equation solveable for 'p'
" " y, x.

Singular soln.

Complementary func
Particular Integrals. CF, PF.
Homogeneous Differential Eqn. (Higher order) ✓

Method of undetermined ~~coeff~~ coeff.
Wronskian ✓

Group Theory

Defn of groups
Permutation group
Dihedral group
Sub group
Commutative group, non-commutative group
Order of the group, element
Cyclic group ~~Pro~~
Even & odd permutation
Alternative group

Defn of group.
When is a group sub-group?

Variable Separation

Q Solve $x\sqrt{(1-y^2)}\,dx + y\sqrt{(1-x^2)}\,dy = 0$

$x\sqrt{1-y^2}\,dx = -y\sqrt{1-x^2}\,dy$

$\Rightarrow \int \frac{x}{\sqrt{1-x^2}}\,dx = \int \frac{-y}{\sqrt{1-y^2}}\,dy$

$= \int \frac{-t\,dt}{t} = \int \frac{z\,dz}{z}$ Let $1-x^2 = t^2$

$= -\int t = \int dz$ $-2x\,dx = 2t\,dt$

$= -t = z + c$ $-x\,dx = t\,dt$

$\boxed{\sqrt{1-x^2} = \sqrt{1-y^2} + c}$ $1-y^2 = z^2$

$-y\,dy = z\,dz$

which is the reqd gen soln.

$c = \pm c$.

Homogeneous Diff Eqn

Q Solve $\frac{dy}{dx} = \frac{-y(x+y)}{x^2}$ — (i) / $x^2 dy = -y(x+y)\,dx$

If same power then homogeneous

This is a HDE

$x^1 y^1 = 1+1 = 2$

y^2

Let $y = vx$.

$\frac{dy}{dx} = v + x\frac{dv}{dx}$.

From (i)

$v + x\frac{dv}{dx} = -y/x - (y/x)^2$

$= -v - v^2$

$= x\frac{dv}{dx} = -2v - v^2$ So

$$\int \frac{dv}{-v(2+v)} = \int \frac{dx}{x}$$

$$\int \frac{\{(2+v) - v\}\, dv}{-2v(2+v)} = \int \frac{dx}{x}$$

$$\Rightarrow -\frac{1}{2}\left[\int \frac{dv}{v} - \int \frac{dv}{2+v}\right] = \int \frac{dx}{x}$$

$$\Rightarrow -\frac{1}{2}\left[\log|v| - \log|v+2|\right] = \log|x| + \log c$$

$$= \log\sqrt{v} - \log\sqrt{v+2} = \log xc.$$

$$= \log\frac{\sqrt{v}}{\sqrt{v+2}} = \log xc.$$

$$\Rightarrow \frac{\sqrt{y/x}}{\sqrt{y/x+2}} = xc.$$

$$\Rightarrow \frac{\sqrt{y}}{\sqrt{y+2x}} = xc.$$

Linear Eqn

Q. The general form of diff a linear diff eqn. of 1st order can be written as

$\frac{dy}{dx} + py = Q$, i) where p & Q are function of x or const.

$\frac{dx}{dy} + px = Q$ ii) where p & x are func of y or const

$$\text{If of (i)} = e^{\int p\,dx}$$

$$\text{(ii)} = e^{\int p\,dy}$$

Q Solve $y^2 + (x - 1/y)\frac{dy}{dx} = 0$

$(x - 1/y)\frac{dy}{dx} = -y^2$

$\frac{dx}{dy} = -\frac{x - 1/y}{y^2} = -\frac{x}{y^2} + \frac{1}{y^3}$

$\frac{dx}{dy} + \frac{1}{y^2}x = 1/y^3 \quad \text{—(i)}$

which is a LDE in x.

$\therefore IF = e^{\int 1/y^2 dy}$

$= e^{\frac{y^{-1}}{-1}}$

$= e^{-1/y}$

y^{-2}

$\frac{y^{-2+1}}{-2+1}$

$\frac{y^{-1}}{-1}$

Multiplying eq (i) by this IF.

$e^{-1/y}\frac{dx}{dy} + e^{-1/y}\cdot\frac{1}{y^2}\cdot x = 1/y^3 \cdot e^{-1/y}$

$= \frac{d}{dy}(x \cdot IF) = 1/y^3 \cdot e^{-1/y}$

$\Rightarrow \int d(x \cdot e^{-1/y}) = \int \frac{1}{y^3} \cdot e^{-1/y} dy$

Let $-1/y = t$

$\frac{1}{y^2}\frac{dy}{dt} = dt$